

1. (15 pts) Let $A = \{\dots, -8, -4, 0, 4, 8, 12, \dots\}$, $B = \{x \in \mathbb{Z} : -2 < x \leq 1\}$, $C = \{1, 2\}$.

(a) Write the set A using set builder notation.

$$A = \{x \in \mathbb{Z} : 4x\}$$

(b) Find $C \times B$.

$$B = \{-1, 0, 1\}, \quad C = \{1, 2\}$$

$$B \times C = \{\{\emptyset\}, \{-1, 1\}, \{-1, 0\}\}, \{0, 1\}, \{0, 2\}, \{1, 1\}, \{1, 2\}\}$$

(c) Is $A \subseteq \mathbb{N}$? Why or why not?

$$A \not\subseteq \mathbb{N} \because -4 \in A \wedge -4 \notin \mathbb{N}$$

2. (10 pts) Construct the truth table for $(\sim Q \wedge P) \implies (P \vee Q)$. Insert any needed columns and be sure to label them.

P	Q	$(\neg Q \wedge P) \implies (P \vee Q)$
T	T	T
T	F	T
F	T	T
F	F	T

3. (10 pts) Write the negation of each of the following statements. You may use words, symbols, or a combination of both. **(Do not prove these statements.)**

(a) If a is even or $5 \nmid b$ then $2a + 4b \leq 25$ and $3 \mid (a - b)$.

$$(2 \mid a \vee 5 \nmid b) \wedge (2a + 4b > 25 \vee 3 \nmid (a - b))$$

(b) $\forall x, y \in \mathbb{Z}, \exists z \in \mathbb{Z}$ such that $((y \mid z) \vee (x + z \text{ is odd}))$.

$$\exists x, y \in \mathbb{Z}, \forall z \in \mathbb{Z} \text{ s.t. } (y \nmid z) \wedge (x + z \text{ is even})$$

4. (10 pts) Prove the following statement:

Suppose $x, y \in \mathbb{Z}$. If $10 \nmid 3xy$ then x is odd or $5 \nmid y$.

Contrapositively, if x is even and $5 \mid y$ then $10 \mid 3xy$.

Proof : $x = 2k \wedge y = 5l$ for some integers k and l .

$$3(2k)(5l) \rightarrow 30kl \rightarrow 10 \mid 30kl.$$

The original statement is true.

5. (15 points) Consider the following new definition:

Definition: An integer x is said to be **fourven** if $x = 4a + 2$ for some $a \in \mathbb{Z}$

(a) Is 10 fourven? Why or why not?

$$\text{Proof : } 10 = 4a + 2$$

$$8 = 4a$$

$$a = 2$$

Therefore, 10 is foureven.

(b) Is 12 fourven? Why or why not?

$$\text{Proof : } 12 = 4a + 2$$

$$10 = 4a$$

$$a = \frac{5}{2}$$

Therefore, 12 isn't foureven.

(c) Let $x, y \in \mathbb{Z}$. Prove that if x or y are even then $2xy + 6$ is fourven.

Case 1 : x is even and y is even.

Proof : $x = 2k \wedge y = 2l$ for some integers k and l .

$$2(2k)(2l) + 6 \rightarrow 8kl + 6 = 4a + 2$$

$$4(2kl + 1) + 2 = 4a + 2$$

This is foureven

Case 2 : x is even and y is odd.

$$\text{Proof : } x = 2k \wedge y = 2l + 1$$

$$2(2k)(2l + 1) + 6 = 4a + 2$$

$$8kl + 4k + 6 = 4a + 2$$

$$4(2kl + k + 1) + 2 = 4a + 2$$

This is foureven

Case 3 : x is odd and y is odd.

$$\text{Proof : } x = 2k + 1 \wedge y = 2l$$

$$2(2k + 1)(2l) + 6 = 4a + 2$$

$$8kl + 4l + 6 = 4a + 2$$

$$4(2kl + l + 1) + 2 = 4a + 2$$

This is foureven.

6. (10 pts) Prove the following statement:

If R is a rational number and $R - 3I$ is an irrational number, then I is irrational.

Proof : Since R is rational, $R - 3I$ should be rational.

Thus I has to be irrational for $R - 3I$ to be irrational.

7. (10 pts) Let $a \in \mathbb{Z}$. Prove that $5a^2 + 7$ is even if and only if a is odd.

*Statement 1 : $5a + 7 = 2k \Rightarrow a = 2k + 1$
for some integer k .*

*Contrapositive : $a = 2k \Rightarrow 5a + 7 = 2k + 1$
 $5(2k) + 7 \rightarrow 10k + 7 \rightarrow 2(5k + 3) + 1$*

The contrapositive is true

*Therefore the statement
is true.*

*Statement 2 : $a = 2k + 1 \Rightarrow 5a + 7 = 2k$
 $5(2k + 1) + 7 \rightarrow 10k + 12 \rightarrow 2(5k + 6)$*

*This proves the statement
true.*

Therefore, this statement is true.